



DPUVDSST4W

This DPU is a simulation of a single track vehicle dynamics for 4-wheels vehicles.

1. Static inputs: (only used in the launch phase)

<i>SetupX, SetupY, SetupZ</i>	Position of the setup point in the global coordinate system [m]
<i>SetupYaw, SetupPitch, SetupRoll</i>	Rotation of the setup point in the global coordinate system [rad]
<i>SetupDone</i>	Flag specifying if DPUSCNX has completely setup the vehicle.

2. Inputs:

<i>AcceleratorPedal</i>	position of the accelerator pedal [0..1]
<i>BrakePedal</i>	position of the brake pedal [0..1]
<i>ClutchPedal</i>	position of the clutch pedal [0..1]
<i>HandBrake</i>	position of the hand brake [0..1]
<i>Ignition</i>	ignition off/on [0/1]
<i>Starter</i>	starter off/active [0/1]
<i>PitchRoad</i>	road pitch angle in rad
<i>RollRoad</i>	road roll angle in rad
<i>Gear</i>	selected gear (manual mode) for automatic transmission: -1 : reverse 0 : neutral 1 : drive 2 : lock gears above 4 3 : lock gears above 3 4 : lock gears above 2 5 : lock gear to 1
<i>minGear</i>	limitation for the automatic transmission; selected gear never is below this gear
<i>SteeringWheel</i>	steering wheel angle in rad
<i>XOffset, YOffset, ZOffset</i>	Additional translation of the vehicle in the simulation's coordinate system during the simulation [m]. This input is computed by DPUSCNX. Default: 0.0 m
<i>ZRoad</i>	height of road [m]
<i>mu</i>	skid number
<i>ACCcontrol</i>	Is ACC in control of acceleration? [0/1]
<i>forcedAx</i>	If ACCcontrol is set, this value [m/s ²] overrides the calculated acceleration, therefore ignoring user inputs via pedals
<i>MOTcontrol</i>	Direct control of motor torque? [0/1]



forcedMmot

If MOTcontrol is enabled, this value [Nm] overrides the calculation of the motor torque by the lookup table given in the configuration file

3. Static parameters:

CfgDriveTrain

configuration file of the drive train

CfgSingleTrack

configuration/parameters of the vehicle dynamics

4. Outputs:

Mmot

static motor torque in Nm

ax

longitudinal acceleration of the vehicle in m/s²

ay

lateral acceleration of the vehicle in m/s²

v

velocity in m/s

v_kmh

velocity in km/h

RPM

rounds per minute of motor 1/min

SelGear

current gear

pitch

pitch angle of vehicle in rad

roll

roll angle of vehicle in rad

yaw

yaw angle of vehicle in rad

vyaw

yaw rate in rad/s

X,Y,Z

absolut position of vehicle in m

consumption

consumption in g/s

5. Dependencies:

This DPU is part of the vehicle dynamic simulation VDS.

6. Functional principle:

Two configuration files are needed which should be located at

LIB/CONFIGURATIONS/VDS

7. DriveTrain

These parameters define the longitudinal dynamics, engine values and gearbox values for the simulated vehicle.

8. Longitudunal dynamic parameters:

vmax1, vmax2

maximum speed of the vehicle in m/s

The vehicle dynamics are limited to this speed. Depending on the parameter *vemin2* either *vmax1* or *vmax2* is used.

ve

road speed [km/h]

vemin2

If *ve* < *vemin2* then *vmax1* is the maximum speed else *vmax2*. This allows for different maximum speeds for cities and motorways for example.

cw

air drag coefficient

A

area of vehicle front in m²

roh

air density (default: 1.2041)

fr

rolling friction



<i>m_v</i>	vehicle mass in kg
<i>m_l</i>	vehicle load in kg
<i>eta</i>	efficiency of drive train [0..1]
<i>rdyn</i>	dynamic wheel perimeter in m
<i>ib</i>	factor to specify how strong braking affects deceleration
<i>ipitch</i>	factor to specify how fast the vehicle reacts on pitch movement
<i>pitch_gain</i>	factor to specify how strong the vehicle is affected by pitch movement and therefor defines the maximum angle
<i>imub</i>	factor to specify the influence of the μ -value of the road on the braking distance
<i>ic</i>	factor to specify the consumption

9. Engine values:

The engines values are defined by an operation characteristic for zero load and full load. This characteristic should contain values at the engine speeds defined by the interval *MotTable_d_Omega* going from engine idle at *MotTable_omega0* to maximum engine speed at *Mot_Table_omega_max*. The number of values derives from the engine speed range and the defined interval. According to the accelerator pedal position (0 for zero load and 1 for full load] and the current engine speed a linear interpolation is done. For non-linear operation characteristics it is possible to define three more arrays *MotTable_Mzw1* to *MotTable_Mzw3* (resembling 0.25, 0.50 and 0.75 accelerator pedal position) for a total of five divisions.

<i>MotTable_omega0</i>	engine idle speed in 1/min
<i>MotTable_omega_max</i>	maximum engine speed in 1/min
<i>MotTable_d_Omega</i>	interval of interpolation points
<i>MotTable_Mmin</i>	engine values for zero load
<i>MotTable_Mmax</i>	engine values for full load
<i>MotTable_Mzw1... 3</i>	optional engine values in between the two above
<i>MotTable_Mmin_Gain</i>	zero load factor
<i>MotTable_Mmax_Gain</i>	full load factor

10. Gearbox parameters:

<i>geartype</i>	1: passenger car gearbox (6 gears) 2: truck gearbox (12 gears) 3: (reserved) for internal use only 4: manual mode (max. 12 gears) with clutch pedal
<i>igR</i>	gear transmission ratio reverse
<i>igN</i>	gear transmission ratio neutral
<i>ig[X]</i>	gear transmission ratio for [X]th gear: [X] range 1 ... 12
<i>igU</i>	reduction gear ratio (truck only)
<i>im</i>	reduction gear ratio engine - gearbox
<i>ik</i>	reduction gear ratio gearbox - Cardan shaft
<i>omega[X][Y]</i>	engine speed threshold for gear change all combinations of [X] [X+1] ([X] range 1 ... 11) and [X][X-1] ([X] range 2 ... 12) must be given



11. SingleTrack

These parameters define the lateral dynamics for the simulated vehicle.

<i>l_F</i>	distance from front axle to the reference point (CoG) in m
<i>l_Ftw</i>	wheel gauge of the front axle in m
<i>l_R</i>	distance from back axle to the reference point (CoG) in m
<i>l_Rtw</i>	wheel gauge of the back axle in m
<i>c_F</i>	suspension roll stiffness of front axle in Nm/rad
<i>c_R</i>	suspension roll stiffness of back axle in Nm/rad
<i>i_SW</i>	steering transmission of steering wheel to wheel angle
<i>m_v</i>	vehicle mass in kg
<i>iroll</i>	factor to specify how fast the vehicle reacts on roll move-ment
<i>roll_gain</i>	factor to specify how strong the vehicle is affected by roll move-ment and therefor defines the maximum angle
<i>UOSteer_v0</i>	vehicle speed in m/s at which the μ -value influences the dynamics
<i>UOSteer_r0</i>	radius in m at which the μ -value influences the dynamics
<i>UOSteer_ratio</i>	ratio of influence of the μ -value on yaw dynamics
<i>UOSteer_mudot_maxpos</i>	limits for the derivation of the μ -value
<i>UOSteer_mudot_maxneg</i>	

12. Example

Drivetrain

VDS DriveTrain

```
{
    # interval
    MotTable_d_omega = 200.0;
    # engine idle speed
    MotTable_omega0 = 500.0;
    # maximum engine speed
    MotTable_omega_max = 3000.0;
    # engine values
    MotTable_Mmin = { -200.0, -140.0, -160.0, -160.0, -200.0, -200.0,
                      -240.0, -280.0, -280.0, -360.0, -360.0, -400.0,
                      -420.0, -420.0, -440.0};
    MotTable_Mmax = { 1000.0, 1080.0, 1200.0, 1280.0, 1300.0, 1300.0,
                      1440.0, 1400.0, 1440.0, 1360.0, 1240.0, 1280.0,
                      1280.0, 1000.0, 100.0};
    # factorisation
    MotTable_Mmin_Gain = 1.0;
    MotTable_Mmax_Gain = 1.0;

    # maximum vehicle speed
    vmax1 = 25; #m/s
    vmax2 = 25; #m/s
    vemin2 = 0;
```



SILAB DRIVING SIMULATION

version Version 7.1 documentation

```
# air resistance coefficient
cw = 0.81;
# frontal area of vehicle
A = 6.02;
# air density
rho = 1.2041; # sea level
# rolling resistance coefficient
fr = 0.01071;
# vehicle mass
m_v = 7800.0;
# vehicle load
m_l = 80.0;
# drivetrain efficiency
eta = 0.81;
# dynamic wheel radius
rdyn = 0.525;
# gearbox type
geartype = 2; # (truck)
# transmission ratios
igR  = -1.0; # reverse
igN  = 0.0; # neutral
ig1  = 5.7; # 1. gear
ig2  = 5.3; # 2. gear
ig3  = 4.7; # 3. gear
ig4  = 4.3; # 4. gear
ig5  = 3.7; # 5. gear
ig6  = 3.3; # 6. gear
ig7  = 2.7; # 7. gear
ig8  = 2.3; # 8. gear
ig9  = 1.7; # 9. gear
ig10 = 1.3; # 10. gear
ig11 = 1.0; # 11. gear
ig12 = 0.7; # 12. gear
igU  = 2.0; # reduction gear ratio
im = 1.0; # engine -> gearbox
ik = 3.5; # gearbox -> Cardan shaft
# thresholds for automatic
omega12  = 1500.0;
omega23  = 1500.0;
omega34  = 1350.0;
omega45  = 1350.0;
omega56  = 1200.0;
omega67  = 1200.0;
```



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```
omega78    = 1200.0;
omega89    = 1200.0;
omega910   = 1200.0;
omega1011  = 1200.0;
omega1112  = 950.0;
omega21    = 1000.0;
omega32    = 1000.0;
omega43    = 1000.0;
omega54    = 1000.0;
omega65    = 900.0;
omega76    = 900.0;
omega87    = 900.0;
omega98    = 900.0;
omega109   = 900.0;
omega1110  = 900.0;
omega1211  = 800.0;
# ratio of brakes
ib = 50000.0;
# pitch ratio
ipitch = 1.0;
# pitch gain
pitch_gain = 0.225;
# influence mu -> braking (0 ... 1)
imub = 0.0;
# consumption factorisation
ic = 0.5;
};
```

Singletrack

VDS SingleTrack4W

```
{
    # distance front axle -> center of gravity
    l_F = 2.47;
    # wheel gauge front
    l_Ftw = 2.0;
    # distance + wheel gauge back axle
    l_R = 1.05;
    l_Rtw = 2.0;
    # roll stiffness front/rear suspension [Nm/rad]
    c_F = 360000.0;
    c_R = 800000.0;
    # steering transmission
    i_SW = 8;
    # vehicle mass
```



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version Version 7.1 documentation

```
m_v = 7800.0;
# roll ratio
iroll = 1.0;
# roll gain
roll_gain = 0.05;
# over-/understeer
U0Steer_v0 = 11.0;
U0Steer_r0 = 10.0;
U0Steer_ratio = 0.5;
U0Steer_mudot_maxpos = 10.0;
U0Steer_mudot_maxneg = -10.0;
};
```

13. SILAB configuration

DPUVDSST4W VDyn

```
{
    Computer = {VDYN};
    Index = 10;
    CfgDriveTrain = "DriveTrain_BMW5.cfg";
    CfgSingleTrack = "SingleTrack_BMW5.cfg";
};
```

Connections =

```
{
    Mockup.AcceleratorPedal -> VDyn.AcceleratorPedal,
    Mockup.BrakePedal -> VDyn.BrakePedal,
    Mockup.SteeringWheel -> VDyn.SteeringWheel,
    Mockup.Gear -> VDyn.Gear
};
```

...

Connections =

```
{
    VDyn.X -> SCNX.X,
    VDyn.Y -> SCNX.Y,
    VDyn.Z -> SCNX.Z,
    VDyn.yaw -> SCNX.yaw,
    VDyn.pitch -> SCNX.pitch,
    VDyn.roll -> SCNX.roll,
    VDyn.X -> SGEView.X,
    VDyn.Y -> SGEView.Y,
    VDyn.yaw -> SGEView.yaw,
    VDyn.Z -> SGEView.Z,
    VDyn.pitch -> SGEView.pitch,
    VDyn.roll -> SGEView.roll,
```



SILAB DRIVING SIMULATION

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```
SCNX.PitchRoad -> SGEView.pitchRoad,  
SCNX.RollRoad -> SGEView.rollRoad,  
SCNX.PitchRoad -> VDyn.PitchRoad,  
SCNX.RollRoad -> VDyn.RollIn,  
SCNX.ZOut -> VDyn.ZRoad,  
SCNX.XOffset -> VDyn.XOffset,  
SCNX.YOffset -> VDyn.YOffset,  
SCNX.ZOffset -> VDyn.ZOffset  
};
```